

CONDENSED OPERATING SEQUENCEIMPORTANT:

The operating sequence described below is applicable only to generators which have been properly installed and are in perfect operating condition.

It is important that the preceding sections of this manual be read before attempting to start-up the generator; in other words, please do not use the following as a shortcut!

5. Set "SOLENOID" selector switch to "OPEN". (if so equipped)

Coil water flows when switch is in "OPEN" position. Plate will not turn on in closed position.

6. Set tank coil tap.

Located in Upper Compartment. Move tap to right for more plate current and to the left for less. See Paragraph 3.5 for proper Adjustment.

7. Push "FILAMENT-ON" pushbutton.

(a) "FILAMENT" pilot light comes on (oscillator filament energized).

(b) "OSC FIL VOLTS" meter reads 22.3 volts.

(c) After 45 seconds red "OVER-LOAD/RESET" pilot light will come on.

(d) Allow 10 to 15 min. warmup for rectifier tube filaments if generator has been off overnight and 30 min. if the air is very cool. Extended warm up not required if equipped with silicon diode rectifier stacks.

3.10 (cont'd)

A. START-UP

	<u>STEP</u>	<u>REMARKS</u>
1.	Connect load coil (with load) to RF terminals.	Do not use teflon tape or pipe compound on coil connections.
2.	Switch on main power at external disconnect box.	
3.	Turn on water supply. Start recirculator pump (if so equipped).	Amber pilot light "RECIRCULATOR ON" will light.
4.	Switch circuit breaker "ON".	
5.	Set "SOLINOID" selector switch to "OPEN". (if so equipped)	Coil water flows when switch is in "OPEN" position. Plate will not turn on in closed position.
6.	Set tank coil tap:	Located in Upper Compartment. Move tap to right for more plate current and to the left for less. See Paragraph 3.5 for proper Adjustment.
7.	Push "FILAMENT-ON" pushbutton.	(a) "FILAMENT" pilot light comes on (oscillator filament energized). (b) "OSC FIL VOLTS" meter reads 22.5 volts. (c) After 45 seconds red "OVER-LOAD/RESET" pilot light will come on. (d) Allow 10 to 15 min. warmup for rectifier tube filaments if generator has been off overnight and 30 min. if the air is very cool. Extended warm up not required if equipped with silicon diode rectifier stacks.

3.10 (cont'd)

START-UP

<u>STEP</u>	<u>REMARKS</u>
8. When the "OVERLOAD/RESET" lamp turns on Push the "OVERLOAD/RESET" button.	"OVERLOAD/RESET" pilot light will come on if time delay has pulled in and plate circuit is ready to be energized.
9. Push green "PLATE ON" pushbutton (if so equipped).	Green "PLATE ON" pilot light comes on. Plate circuit is now energized.
10. Push green "RF POWER-ON" pushbutton.	Green "RF POWER" pilot light now comes on. R. F. plate power is now on.
11. Adjust "POWER CONTROL" knob.	Power output can be set to desired level between minimum and maximum. As control advanced (C.W.), "Plate Kilovolts", "Grid Amperes", and "Plate Amperes" meter reading increase to maximum.

B. SHUTDOWN

1. Push red "R. F. POWER" stop pushbutton.	Green "RF POWER" pilot light goes out. RF power and H. V. plate supply to RF section off.
2. Push red "PLATE-STOP" pushbutton (if so equipped).	Plate contactor de-energizes and H. V. supply is off. Green "Plate" pilot goes out. Green "FILAMENT" pilot light goes out. Oscillator filament power is off.
3. Push red "FILAMENT-STOP" pushbutton.	

3. (cont.)

SHUTDOWN

STEP

REMARKS

- | | |
|---|---|
| 4. Turn off recirculator pump front panel selector switch and turn off water supply to recirculator. | Amber pilot light goes out on recirculator control panel. |
| 5. Remove line input power by external disconnect switch or turn main circuit breaker "off" (located bottom left front control panel) | Remove all power from generator circuits.
Shutdown complete. |

NOTE:

Wait five minutes after Step 3 (turning off filament) before turning off water to allow oscillator tube to cool properly.

SECTION 4 MAINTENANCE

4.1 ROUTINE MAINTENANCE

As with all industrial equipment the ideal schedule for periodic maintenance is a function of usage and environment. An initial inspection three months after installation with inspections every six months thereafter is suggested.

Inspection should include:

- (a) A check of hoses for condition and leaks.
- (b) Removal, washing and replacement of the air filters in side panels if so equipped.
- (c) A check for dirt and dust; be sure to clean rectifier and other high voltage components if dust has accumulated.
- (d) Removal of oscillator tube, check anode for scaling and clean if required.
- (e) Make sure tube terminal connections are clean and tight.
- (f) If equipped with coolant recirculator, check reservoir level. Fill to within 2 inches of tank top when pump is off if required. The quality of the recirculated water will eventually deteriorate and so it should periodically be drained and refilled with distilled water. Minimum water quality would be minimum resistivity of 5000 Ω - centimeters and a maximum total hardness of 170 parts per million (or 10 grains per gallon.)

4.2 RP TROUBLE SHOOTING GUIDE FOR SCHEMATIC DIAGRAM T-1001-237

<u>TROUBLE</u>	<u>POSSIBLE CAUSE</u>	<u>CHECK*</u>
1. No Filament Voltage.	(a) External supply fuse open, (b) Circuit breaker open, (c) Open fuse, (d) Missing supply voltage phase, (e) Poor cooling, check red fault lamps in "PROTECTIVE SYSTEM" block on front panel, (f) Recirculator pump off (If so equipped).	CB1 F1, F2, F3, F4
2. "OVERLOAD RESET" lamp does not light 45 seconds after "FILAMENT ON" lamp turns on.	(a) Look for red indicator lamp in "PROTECTIVE SYSTEM" block on front panel, (b) Defective overload reset relay, (c) Open time delay relay.	K8 K6
3. "FILAMENT" lamp comes on but will not stay on when "FILAMENT ON" button is released.	(a) Missing supply voltage phase, (b) Missing supply voltage phase relay inoperative.	K3
4. Plate contactor fails to close.	(a) Time delay relay not closed, (b) Make sure "OVERLOAD RESET" button was pressed, "OVERLOAD RESET" lamp is out, and "PROTECTIVE SYSTEM" lamps are out, (c) Defective plate contactor.	K6 K9
5. "OVERLOAD RESET" lamps come on when "RF POWER" button is pushed.	(a) Work coil overcoupling, move tank coil tap to lower number or increase spacing between coil and work or reduce number of coil turns.	

<u>TROUBLE</u>	<u>POSSIBLE CAUSE</u>	<u>CHECK*</u>
	(b) "GRID CONTROL" setting is low, increase,	
	(c) Open Tank Circuit,	
	(d) High resistance load coil connection do not use pipe joint compound or teflon tape on coil connections,	
	(e) Shorted plate blocking capacitor,	C6
	(f) Shorted DC Filter capacitor,	C4, C5
	(g) Shorted Grid Capacitor,	C10
	(h) Shorted tank capacitor,	C7
	(i) Defective rectifier tube,	V5, V6, V7
	(j) Defective thyatron tube,	V1, V2, V3
	(k) Defective oscillator tube.	V4
6. Plate current present but no grid current.	(a) Same as Symptom 5 above.	
7. Low RF output, low plate current reading.	(a) Defective Rectifier tube,	V5, V6, V7
	(b) Defective thyatron tube,	V1, V2, V3
	(c) Defective oscillator tube.	V4
8. No plate current but plate contactor operating.	(a) Open circuit in DC power supply,	
	(b) Defective oscillator tube,	V4
	(c) Defective T-660 Thyatron regulator, see instruction manual (No. 13190002).	A1, A2

*NOTE: The letter references in this column are reference designations for components in Schematic Diagram No. T-1001-237.

2. DIMENSIONS APPROX

GENERATOR-REAR

8

1 1/2

1" NPT

1" NPT

GRD.

IN OUT

1/4"

6 1/2

3 1/2

8

2 PANELS
BOTH SIDES
(REMOVABLE)

RF OUTLET- FLARED
MALE CONNECTOR
1/4 NPT x 3/8 TUBE

*

____ CONDUCTOR SIZE (AWG)

COOLING WATER SUPPLY

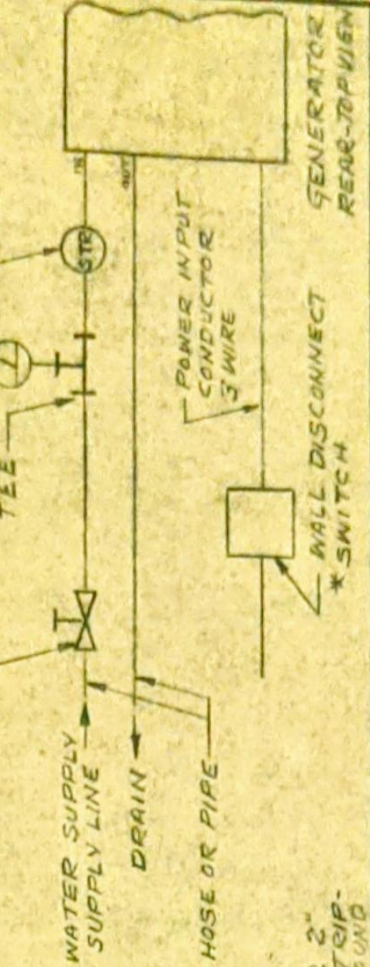
_____ G.P.M. _____ TEMP. (MAX)

_____ P.S. 1. MAX.

P.S. 1. MIN.

✓ PRESSURE GAUGE

WATER STRAINER



GRD. - #6 WIRE OR 2"
WIDE COPPER STRIP-
TO BUILDING GROUND

NAME

NAME _____
INSTALLATION DIAGRAM

GENERATOR FI

MODEL T-7.5 TO T-20 KVA

LEFT

HIGH FREQUENCY LABORATORIES, INC.
59-21 QUEENS-MIDTOWN EXPRESSWAY
MASPETH, NEW YORK CITY

DRAWING NO.

REV

9570146

MATERIAL

FINIS

SHEET 1 OF 1

SCALP

OF

UNLESS OTHERWISE SPECIFIED
DIMENSIONS IN INCHES

TOLERANCE ON:
FRACTIONS DECIMALS ANGLES

— 卅 —

DRAWN BY	DATE
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W. Johnson 1-16-77

CHECKED *[Signature]* DATE 6-18-7

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ENGINEER
B. J. ... 6-18-23
DATE

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APPLICANT	APPLICATION	DATE	STATUS	REMARKS
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APPLICANT	APPLICATION	DATE	STATUS	REMARKS
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LEPEL HIGH FREQUENCY LABORATORIES Inc.

GENERATOR INSTALLATION SPECIFICATIONS

NOTE: 1

NOTE: 2

NOTE: 3

COOLING WATER NOTES

GENERATOR MODEL	POWER INPUT KVA	PHASE	INPUT LINE VOLTS	FULL LOAD CURRENT AMPS	ELECTRICAL FUSE AMPS	SERVICE CABLE AWG	MCM	WATER REQ.** GPM	WATER FITTING NPT*** INLET	OUTLET
T-1-1	2.2	1	115	21	40	8		1	*1/4	* 1/4
T-2-5	5.5	1	230	24	40	8		2	*1/2	*1/2 *1/4
			460	12	25	10				
T-5-3	12.7	3	230	32	45	8		4	*1/2	*1/2 *1/4
			460	16	30					
			550	13.5	25	10				
T-7.5-3	16.7	3	230	42	60	6		4	1/2	*1/2 3/8 3/8
			460	21						
			550	17.5	30	10				
T-10-3	25.4	3	230	64	110	2		8	1	1-1/4
			460	32	60	6				
			550	27						
T-15-3	32.8	3	230	82	125	1		10	1	1-1/4
			460	41	80	4				
			550	34						
T-20-3	46	3	230	116	175	00		10	1	1-1/4
			460	58	80	4				
			550	47						
T-25-3	58	3	230	144	250	250		14	1	1-1/4
			460	72	125	1				
			550	60						
T-30-3	68	3	230	170	250	250		14	1	1-1/4
			460	85	125	1				
			550	71						
T-35-3	80	3	230	200	300	350		22	1	1-1/4
			460	100	175	00				
			550	84	125	1				
T-40-3	89	3	230	224	350	500		22	1	1-1/4
			460	112	175	00				
			550	94						
T-50-3	120	3	230	300	450	700		22	1	1-1/4
			460	150	250	250				
			550	120	175	00				
T-60-3	143	3	230	360	500	—		22	1	1-1/4
			460	180	250	—				
			550	150						
T-80-3	192	3	230	480	800	—		30	1	1-1/4
			460	240	400	—			****	****
			550	200	350	—				
T-100-3	240	3	230	600	1000	—		35	1	1-1/4
			460	300	450	—			****	****
			550	250	400	—				

* Denotes water hose nozzle.

** The inlet pressure must fall between 40 PSI, the minimum pressure required to maintain the specified flow rate shown, and 65 PSI, the maximum allowable pressure.

If the inlet pressure falls below 40 PSI, install a booster pump in the water supply line.

If the inlet pressure exceeds 65 PSI install a pressure regulator in the water supply line.

Water inlet temperature must not exceed 80°F (70°F if the generator is equipped with the sealed cabinet air cooling).

When a generator is supplied with a water recirculator utilizing a water to water heat exchanger, the consumption of city water is automatically adjusted to the varying cooling requirements of the generator.

When a generator is so-equipped, the water supply requirements increase by 3 GPM for generators from T-10-3 to T-40-3 and by 5 GPM for generators from T-50-3 to T-100-3.

*** To insure an adequate flow, the sizes of the water supply and drain lines should equal or exceed those of the 'inlet' and 'outlet' fitting sizes.

Gravity outlet drains are recommended.

If the use of an overhead drain is unavoidable, install a check valve in this line NEVER a manually operated valve.

**** 1 1/2" when a water recirculator is used.

ELECTRICAL NOTES

- For information on generator models not listed please contact Lepel High Frequency Labs. Inc.
- The listing of 'input line volts' are nominal and Lepel generators are designed to operate over the following voltage ranges.

Nominal

Range

115 Volts	105 Volts to 125 Volts
230 Volts	200 Volts to 250 Volts
460 Volts	400 Volts to 500 Volts
550 Volts	500 Volts to 600 Volts

- 'Electrical Service' listings are based on the maximum line current requirements for the lowest acceptable line voltage.

In general, Lepel line circuit breakers are rated at 125% of this maximum line current.

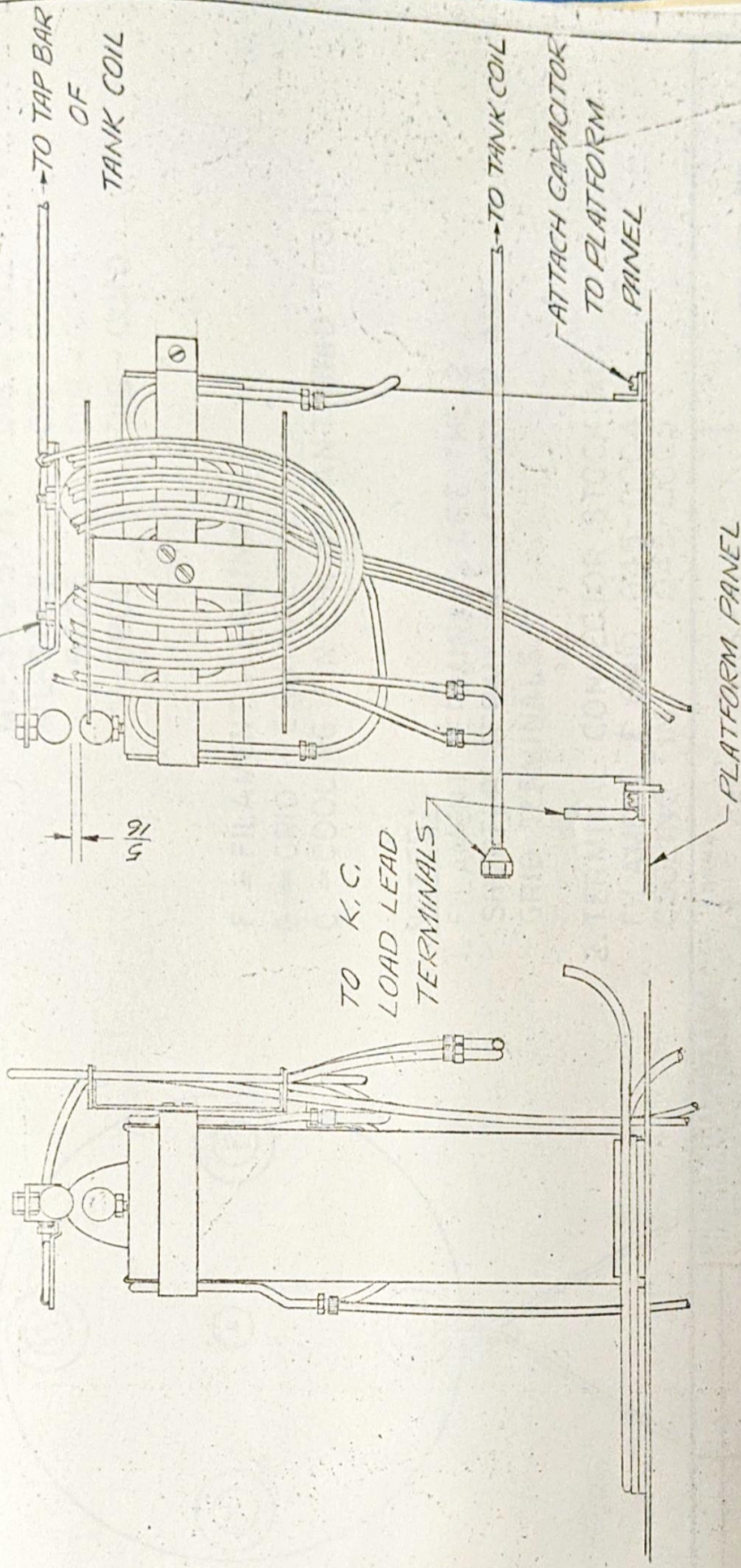
The cable sizes listed conform to the National Electrical Code, Table 310-14 for copper conductors, rated at 75°C. However, it is recommended, that the user consult the local electrical code requirements for conductor ampacities.

DWG NO. 9570041

REV. 6-27-74

MARKED WITH CORRESPONDING
NUMBERS

ATTACH PLATE TO TOP OF CAPACITOR



USED WITH:		MATERIAL:		SCALE		ASSEMBLY D'W'G. No.	
				NOT TO SCALE			
ENG.		DRAFT		9-10-62		9-10-62	
CHECK		H-2		9-11-62		7392-2 B	
DATE		BY		REVISION		D'W'G. No.	
11/15/63		11/15/63		11/15/63		11/15/63	
B		CHANGED FROM RUBBER		11/15/63		11/15/63	
A		REMOVED HOSE COUPLING		11/15/63		11/15/63	
LET.		REVISION		17710		17710	

LEPEL

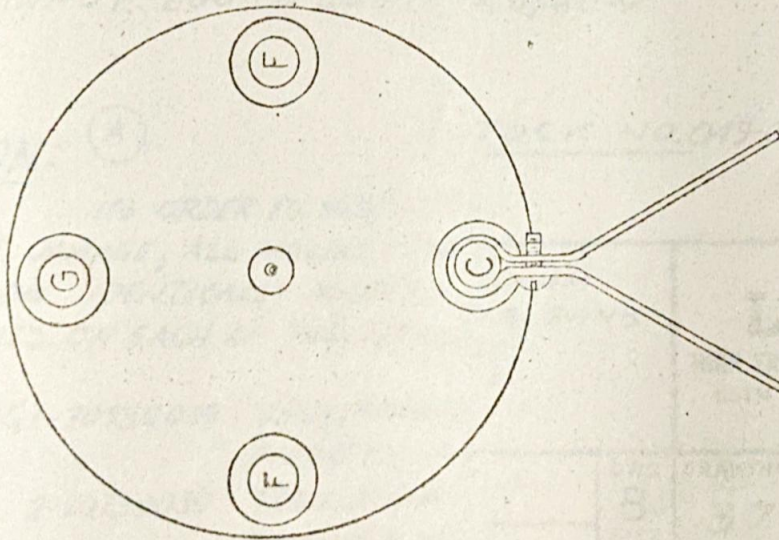
HIGH FREQUENCY LABORATORIES, Inc.
54-18 37TH AVE. WOODSIDE, N. Y.

NAME
INSTALLATION DIAGRAM
TANK CAPACITOR

17710

AND SHALL NOT BE SUBMITTED TO OUTSIDE PARTIES FOR EXAMINATION WITHOUT OUR KNOWLEDGE AND CONSENT; IT SHALL BE USED AS A MEANS OF REFERENCE TO WORK FURNISHED OR SUGGESTED BY THIS COMPANY ONLY.

TOP VIEW OF TUBE



INSTALLATION CONNECTIONS FOR:

TYPE	STOCK NO.
ML-356/5771	018 - 0012
ML-5668	018 - 0009
5606	018 - 0005
6800	018 - 0010

- F = FILAMENT TERMINALS
- G = GRID TERMINAL
- C = COOLING FIN (ON VACANT GRID TERM)

NOTES:

- FILAMENT TERMINALS ARE THE 2 SHORTER TERMINALS - OTHER 2 ARE GRID TERMINALS.
- TERMINAL CONNECTOR STOCK NO. FILAMENT & GRID 045-0004 COOLING FIN 045-0005

UNLESS OTHERWISE SPECIFIED DIMENSIONS IN INCHES		TOLERANCE ON: FRACTIONS DECIMALS ANGLES		± ± ±	
DRAWN BY	DATE	CHECKED		DATE	
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APPROVED FOR MFG		ENGINEER		DATE	
		R.B.		8.22.63	
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OSCILLATOR TUBE TERMINAL DIAGRAM

MATERIAL	FINISH

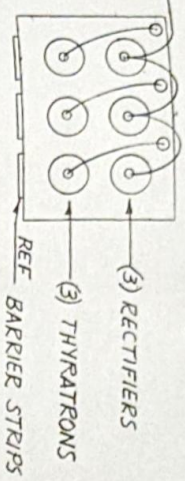
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SHEET 1 OF 1	
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LEPEL
HIGH FREQUENCY LABORATORIES, INC.
55TH STREET AND 37TH AVENUE
WOODSIDE, N. Y.

NOTE: (B)

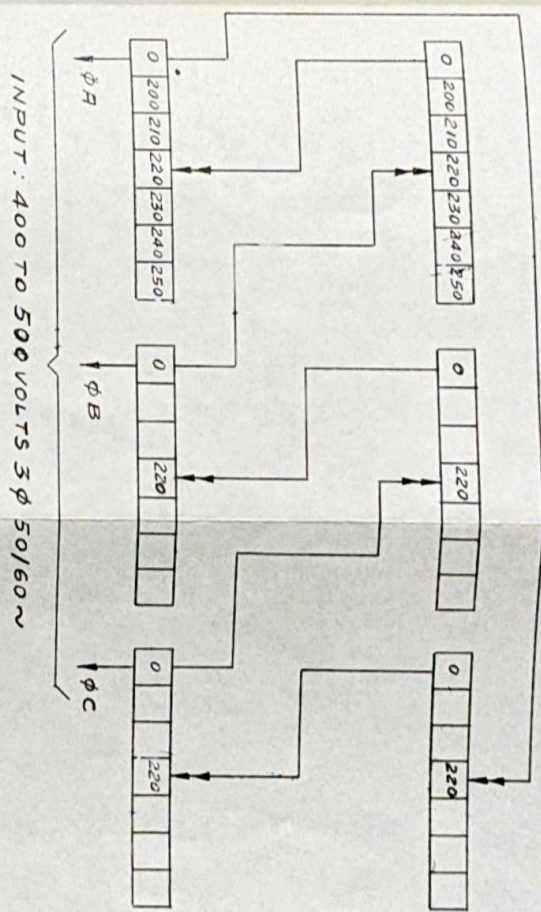
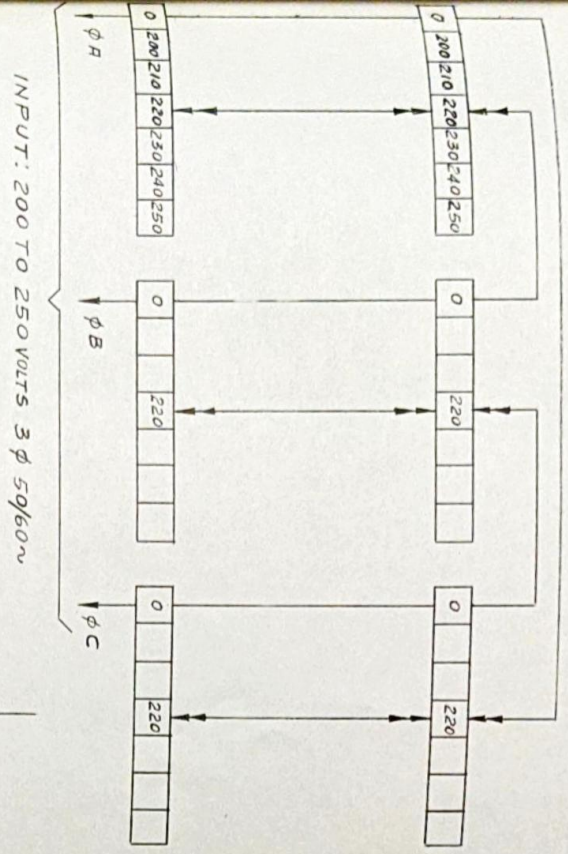


NOTE:

1- RECTIFIER & THYRATRON ANODE CONNECTIONS MUST BE MADE AS DIAGRAMMED

REVISIONS

SYM	ZONE	DESCRIPTION	DATE	APPROVAL
A		REMOVED STK NO. 049-0205, ADDED STK NO. 049-0205, ADDED "CAUTION"	11/11/66	14C
B		ADDED PLAN VIEW OF TUBES & NOTE NO. 1	1/29/73	14C



CAUTION: (A)

IN ORDER TO PROTECT THE TRANSFORMER AGAINST DAMAGE, ALL MOVABLE TAPS MUST BE OPERATED ON IDENTICALLY MARKED VOLTAGE TERMINALS ON EACH OF THE TERMINAL BOARDS

THIS DIAGRAM USED FOR LEPEL STOCK NO. 049-0205, 049-0206

REF DWG 1-20730029 - TRANSFORMER RECTIFIER FILAMENT - UNITIZED (730, 40, 50) 2-20730030 - TRANSFORMER, RECTIFIER FILAMENT - UNITIZED (755, THE T25)

20730029		755 TRANSFORMER		DATE		DRAWN BY		UNLESS OTHERWISE SPECIFIED		NAME		DWG		REV	
20730029		755 TRANSFORMER		DATE		G-T-G		DIMENSIONS IN INCHES		UNITIZED RECTIFIER BANK		B		3730005	
NEXT ASSY		USED ON		DATE		6-7-66		TOLERANCES IN INCHES		LINE VOLTAGE		SCALE		SHEET 1 OF 1	
APPLICATION		ENGINEER		DATE		6-7-66		FRACTIONS DECIMALS		ADJUSTMENT		FINISH			
		W. J. G. - 6-7-66		APPROVED FOR MFG		DATE		ANGLES		HIGH FREQUENCY LABORATORIES, INC.					
										WOODSIDE, N. Y.					

INSTRUCTION MANUAL

THE LEPEL T-660 THYRATRON REGULATOR

WITH AUTO-MANUAL REMOTE CONTROL

(MANUAL NO. 13190002)

Advantages and Special Features of the T-660

- 2-1) Smooth continuous control of the power supply output from zero to 100%.
- 2-2) The output is essentially linear and proportional to the input control signal.
- 2-3) High gain.
- 2-4) Precise and synchronized pulses to grids of the thyristors permit accurate and repetitive control.
- 2-5) Low voltage consumption.
- 2-6) Superior reliability - all solid state.
- 2-7) Insensitive to the average value of input signals. Rejection of both slow and fast extraneous short term disturbances have no effect on the output "I.A." or false triggering.

LepeL HIGH FREQUENCY LABORATORIES, Inc.

THE LEPEL T-660 THYRATRON REGULATOR

1. Description

The Lepel T-660 Thyatron Regulator is a control system which permits precise and reliable control of the high voltage D.C. output from a three phase power supply. The latter furnishes the D.C power for Lepel High Frequency Induction Heating generators.

The Lepel T-660 incorporates a magnetic amplifier which generates narrow pulses synchronized 120° apart which are used for triggering the grids of thyatron tubes. The sharp rising wave front of each pulse minimizes any chance of misfiring of the thyatrons.

2. Advantages and Special Features of the T-660

- 2-1) Smooth continuous control of the power supply output from zero to 100%.
- 2-2) The output is essentially linear and proportional to the input control signal.
- 2-3) High gain
- 2-4) Precise and synchronized pulses to grids of the thyatrons permit accurate and repetitive control.
- 2-5) Line voltage compensation.
- 2-6) Maximum reliability - all solid state.
- 2-7) Responds only to the average value of input signals. Noise spikes and other extraneous short term disturbances have no effect on the output "i.e." no false triggering.
- 2-8) Completely packaged in two modules making installation and service simple.

- 2-9) Internal electronic keying.
- 2-10) Internal bias supply for thyratrons.
- 2-11) Internally regulated signal supply (0 to 5 MADC)
- 2-12) Short duration grid pulse minimizes thyatron grid dissipation.
- 2-13) Flexible control system permits application of most types of instrument control.
- 2-14) Facilities provided for external bias or feedback systems.
- 2-15) Electrostatic shielding and grounding of low and high voltage systems.

3. Specifications

- 3-1) The Lepel T-660 Thyatron Regulator consists of a low voltage assembly (T-660A) and a high voltage assembly (T-660B)

Dimensions: T-660A (LxDxH) $8\frac{1}{2}" \times 5-1/8" \times 4\frac{1}{2}"$
 T-660B (LxDxH) $10" \times 4-5/8" \times 8\frac{1}{2}"$

A 84 inch interconnecting cable is furnished for electrically connecting the T-660A to the T-660B. The cable length is not critical and may be increased upon request.

- 3-2) Power Input: 220/440 Volts Three Phase 60 cycles (50 cycles available)

The change from 220 volt to 440 volt operation and vice versa is made by changing a pre-wired fanning strip. Both 220 and 440 volt fanning strips may be furnished upon request.

- 3-3) Output: 350 volt, three phase pulses are generated in the T-660 which are fed to the grids of three H.V. thyratrons. The latter are part of the three phase full wave high voltage rectifier bank of the Lepel generator.

- 3-4) Response Time of the T-660: (Without D.C. Filtering in the power supply)
 Thirty (30) milliseconds or less.

- 3-5) Range of power control: Zero to 100%.

3-6) Generator Range: T-5 to the T-100 generators
(5 KW to 100 KW generators)

3-7) Insulation:- The high voltage section is insulated to accommodate D.C. output voltages to 15 K.V.
The control windings are insulated for 2000 Volts D.C.

3-8) Types of Thyratrons controllable with the T-660.

3-8.1) 5869/9950

3-8.2) 678

3-8.3) 5563A

3-8.4) 5870/9951

3-8.5) 8270

3-8.6) 6786

3-9) Internal bias supply is provided with voltage adjustments to operate with the thyratrons mentioned in item (8).

3-10) Electronic keying (pulsing)

The T-660 circuitry permits rapid keying of the D.C. power supply output. The thyatron grid voltages are removed and the negative bias supply (item 9) blocks the conduction of the thyratrons. The latter, when blocked, deliver no D.C. output voltage to the generator.

3-11) Line Voltage Compensation:

The T-660 is equipped with an internal feedback loop which automatically compensates for line voltage variations. A $\pm 10\%$ line voltage variation is compensated to a $\pm 2\%$ variation at manual output control adjustments up to 85% of the power supply D.C. output voltage.

3-12) Control System Features of the T-660

a) Manual control with a 1000 ohm 2 watt potentiometer using the internal voltage regulated power supply.

Full control range is obtained with 2.5 milliampere turn signal inputs.

For convenience of the user, Lepel can furnish at an additional cost the "Auto-Manual" control panel. The latter is connected to the T-660A to provide switching between "Manual" and "Automatic" operation.

Refer to Drawing #9670004

Switch S1 changes from "Manual" to "Automatic" operation.

Switch S2 changes from the "0 to 5 Ma" DC range to "4 to 9 MA" range.

Input jacks (not indicated) to the main control winding and the auxiliary winding.

4. PHASING PROCEDURE

* 4-1) Set the "Manual-Auto" toggle switch to the "MANUAL" position.

* 4-2) Set the Control current toggle switch to 0 to 5 MADC.

4-3) Set the Manual output control 1000 ohm potentiometer to minimum or to the extreme counter clockwise position.

4-4) When the T-660 thyatron regulator is properly phased with respect to the main plate transformer, zero to maximum D.C. voltage control will be obtained by adjusting the Manual output control from minimum to maximum positions.

At maximum, rated kilovolts will be indicated on the D.C. plate kilovoltmeter. If the regulator is improperly phased, the Manual output control will not control from zero to maximum, when adjusted over its range.

4-5) Phasing is accomplished by rotating the three phase input wires connecting to terminals (1) (5) and (9) of the low voltage section T-660A. There will be six combinations to cover phasing.

In each test, it is recommended that all power be removed from the equipment. In checking each combination, the output control should be adjusted to see if the regulator is functioning (Refer to item (4)). If the six combinations are checked and correct phasing is still not obtained reverse two of the main lines to the generator and repeat item (5).

* For "Auto-Manual Remote Control Only."

SUGGESTED MEASUREMENTS TO DETERMINE DEFECTIVE SECTIONS OF T-660

1) LOW VOLTAGE SECTION T-660-A

Failure of internal components of low voltage unit can be ascertained by removing interconnecting cable fanning strip from terminals #16 to #28 and by checking D.C. voltages from terminal #15 (Ground) to terminals #16 thru #28. (Output Control Set At Zero).

From terminal	#15	to terminal	#16	1.6	Volts	D.C.
"	"	"	#17	30	"	"
"	"	"	#18	1.6	"	"
"	"	"	#19	30	"	"
"	"	"	#20	1.6	"	"
"	"	"	#21	30	"	"
"	"	"	#22	20	"	"
"	"	"	#23	0	"	"
"	"	"	#24	36	"	"
"	"	"	#25,28	0	"	"
"	"	"	#26,27	0	"	"

2) HIGH VOLTAGE SECTION T-660-B

To check high voltage unit remove wires at terminals K, G1, G2, G3 and measure a D.C. voltage of 125 volts (235 volts if using 9951 or 8270 thy. tubes) from terminal "K" to terminals G1, G2 and G3.

Between terminals #29 and 30 and A.C. voltage of 2.5 volts should be read. Voltage reading between terminals #29 and 31 should be 5 volts A.C.

3) IMPORTANT: All measurements are made with Plate Power Off!

4) NOTE: When using the type 6786 thyatron, use connections #29 and #31. The bias voltage from the grids G1, G2 and G3 to K will read (minus) -125 Volts DC.

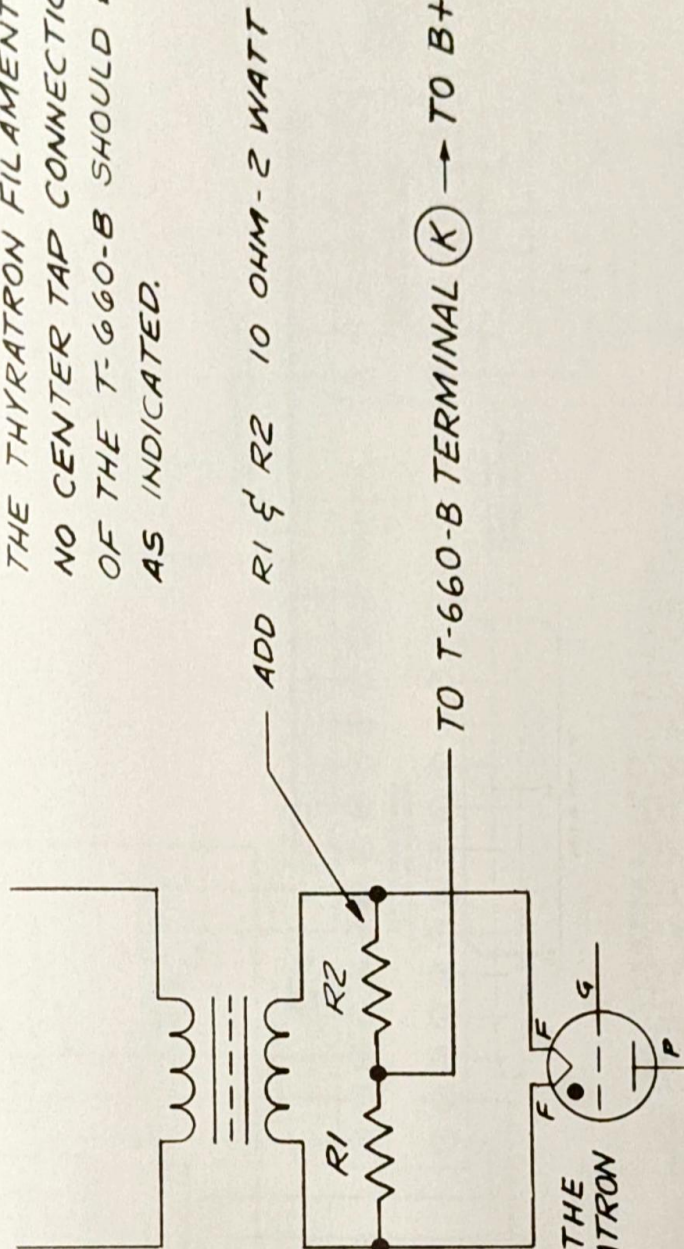
TROUBLE SHOOTER'S GUIDE T-660 THYRATRON REGULATOR

Important: All measurements are to be made with filaments "ON" only.
Tank coil panel interlock must be open. PLATE POWER MUST BE OFF.

TROUBLE	POSSIBLE CAUSE	CHECK
Very rough power regulation, thyatron tubes will not fire with power control set at zero.	a. Generator is out of phase.	Reverse any two incoming lines to generator.
No R.F. power output but plate contactor operating.	a. Power control potentiometer defective.	Check 1 K ohm potentiometer between terminal 22 and 28.
R.F. power goes on at almost full power with no power regulation.	a. Open circuit in 5 volt filament supply to high voltage unit. b. No D.C. voltage applied to grid of thyatrons.	Check for 5 volts A.C. terminals 29 and 31 Check for 125 volts D.C. (235 volts D.C. if using 9951, 8720 thy.) from terminal "K" to terminals G1, G2 and G3
Erratic power control 1 thy. tube will fire with power control set at zero.	a. Phase A,B or C on low voltage unit open. b. Grid of 1 thy. tube open. c. Shorted grid to filament 0.02 MFD capacitor (Replace capacitor)	Check for 220 or 440 volts A.C. between terminals 1 and 5, 5 and 9 on low voltage unit. Check for D.C. grid voltage between terminal "K" and G1, G2 and G3. Check capacitor.
Erratic power regulation, unit will activate overload relays	a. Open grid to filament 0.02 MFD capacitor.	Check capacitor.

NOTE:

1. IN OLDER GENERATOR MODELS, WHERE THE THYRATRON FILAMENT WINDING HAS NO CENTER TAP CONNECTION, TERMINAL "K" OF THE T-660-B SHOULD BE CONNECTED AS INDICATED.



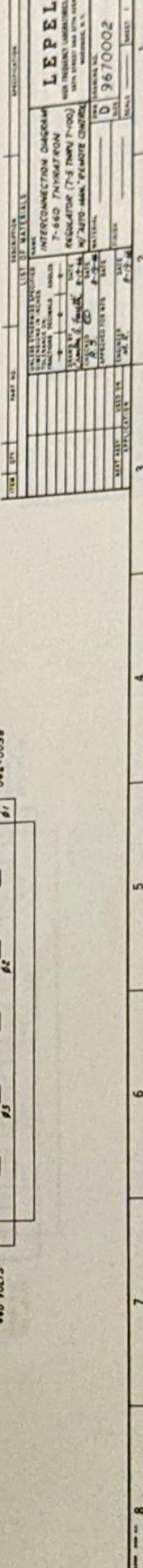
ANY ONE OF THE
THREE THYRATRON
TUBES ONLY.

	UNLESS OTHERWISE SPECIFIED DIMENSIONS IN INCHES TOLERANCE ON: FRACTIONS DECIMALS ANGLES	DATE 6/20/67
	DRAWN BY Fred Bystsch	DATE 6-20-67
	CHECKED All Personnel	DATE 6-20-67
	APPROVED FOR MFG	DATE
	ENGINEER	DATE
NEXT ASSY	USED ON	

NAME SCHEMATIC DIAGRAM T-660 CONNECTION TO OLD TYPE GENERATORS	
MATERIAL	DWG A
FINISH	SIZE
SCALE	

LEPEL
HIGH FREQUENCY LABORATORIES, INC.
55TH STREET AND 37TH AVENUE
WOODSIDE, N. Y.

DRAWING NO. 19140210	REV
SHEET / OF /	



2. RESISTOR VALUES TO CORRESPOND WITH TEMPERATURE CONTROLLER REQUIREMENTS. (NOT LESS THAN 100 OHMS).
3. TERMINAL IS AND CASE OF LOW VOLTAGE UNIT ARE COMMON AND MUST BE GROUNDING. NEVER REMOVE GROUND FROM TERMINAL IS.
4. USE OF MFD. 1600 CAPACITORS (STOCK NO. 007-0072) ACROSS EACH THYRATRON GRID AND FILAMENT. (CONNECTED AT SOCKETS).
5. FOR INSTRUMENT CONTROL, REFER TO TECHNICAL MANUAL # 13190002.
6. GENERATORS HADSE THYRATRON FILAMENT WINDINGS ARE NOT EQUIPPED WITH CENTER TAP CONNECTIONS, REFER TO DMC 1194020.

A- REMOVE, INSULATE AND TAPE WIRES L1,L2,L3 LOCATED ON PRIMARY OF CONTROL TRANSFORMERS.

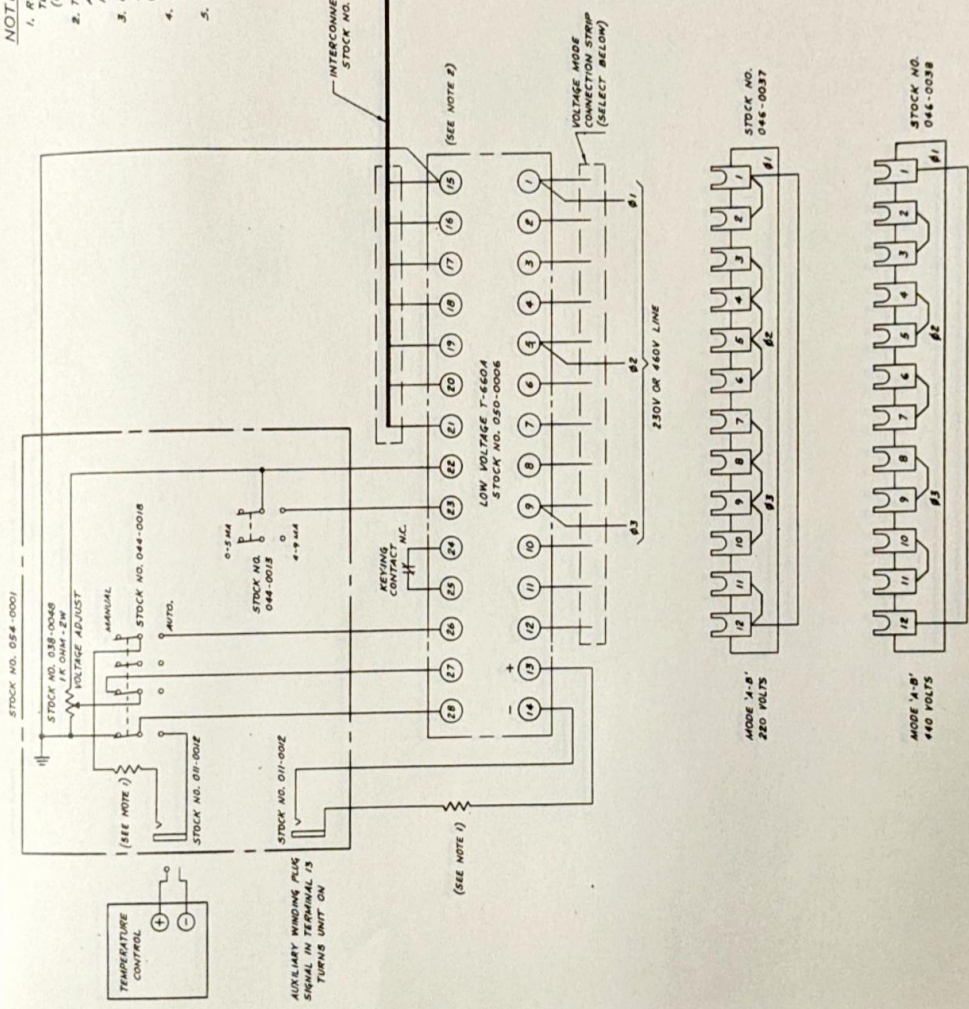
B- REMOVE RESISTORS FROM THYRATRON TUBE SOCKETS.

C- REMOVE EXISTING CAPACITORS FROM THYRATRON TUBE SOCKETS AND REPLACE WITH CAPACITORS AS DESCRIBED IN NOTE 3. (SEE BELOW)

D- FOR SETUP AND PHASING PROCEDURE, REFER TO TECHNICAL MANUAL # 13740002.

• WHEN USING 9950 OR 678 THYRATONS
USE TERMINAL 31 INSTEAD OF 10.

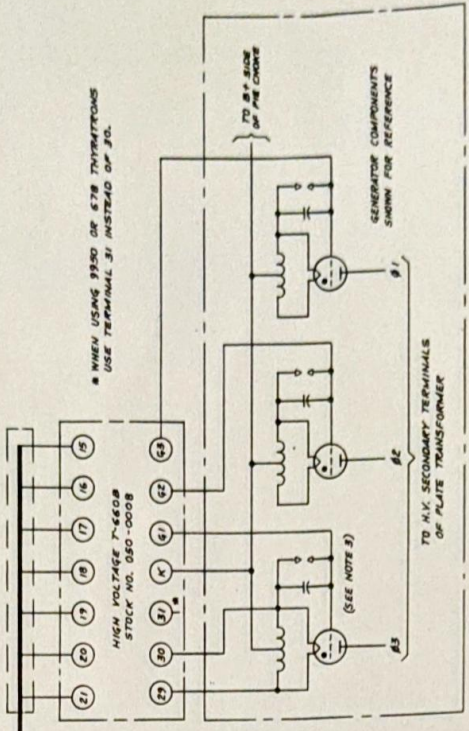
MODE 'A-B'
440 VOLTS



NOTES:

1. RESISTOR VALUES TO CORRESPOND WITH TEMPERATURE CONTROLLER REQUIREMENTS. (NOT LESS THAN 100 OHMS).
2. TERMINAL IS AND CASE OF LOW VOLTAGE UNIT ARE COMMON AND MUST BE GROUNDED. NEVER REMOVE GROUND FROM TERMINAL IS.
3. USE .02 MFD, 1600V CAPACITORS (STOCK NO. 307-0074) ACROSS EACH THYRATRON GRID AND FILAMENT. (CONNECT AT SOCKETS).
4. FOR INSTRUCTIONS, REFER TO TECHNICAL MANUAL # 13910002.
5. GENERATORS MOOSE THYRATRON FILAMENT WINDINGS ARE NOT EQUIPPED WITH CENTER TAP CONNECTIONS. REFER TO DING #140890.

- CONVERSION TO T-660
- A- REMOVE, INSULATE AND TAP WIRE L1,L2,L3 LOCATED ON PRIMARY OF CONTROL TRANSFORMERS.
- B- REMOVE RESISTORS FROM THYRATRON TUBE SOCKETS.
- C- REMOVE EXISTING CAPACITORS FROM THYRATRON TUBE SOCKETS AND REPLACE WITH CAPACITORS AS DESCRIBED IN NOTE 3. (SEE BELOW)
- D- FOR SETUP AND PHASING PROCEDURES, REFER TO TECHNICAL MANUAL # 1390002.

[illegible]

8

THE PRINCIPLES AND APPLICATION OF INDUCTION HEATING

PRINCIPLES

When alternating electric current flows in a conductor, a symmetrical electromagnetic field is established around the conductor. If the conductor is formed into a loop or a coil, the alternating magnetic field is intensified; the actual intensity being determined by (a) the magnitude of the current flowing through the coil, (b) the number of turns in the coil. This magnetic field is distorted, as shown, in figure 1.

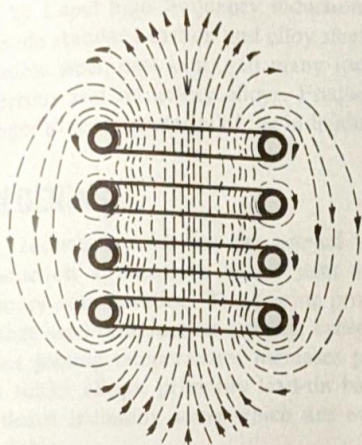


Figure 1

Circulating currents will flow in any electrically conductive material placed in this magnetic field, the value of the 'induced' current being determined by the strength of the alternating magnetic field and the spacing between the workpiece and the coil. Figure 2.

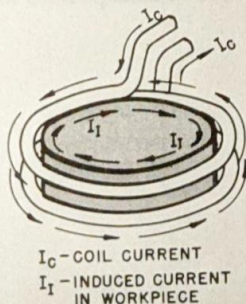


Figure 2.

The resistance offered by the workpiece to the flow of the induced current produces heat in it proportional to the electrical resistivity of the workpiece and to the square of the current flowing. Thus, any metal part or other conductive material placed within a coil energized with alternating electric current heats without physical contact between the part and the coil . . . i.e., it heats by INDUCTION. High coil-current and close spacing between the coil and workpiece produce maximum heating.

SURFACE HEATING

When high frequency current is used for induction heating, the induced current is a maximum at the surface and rapidly decreases toward the interior of the part. This effect is more pronounced at higher frequencies, and permits selective heating of the surface.

In practice, the effective depth of heating is determined by thermal conduction of heat from the surface to the interior of the part, as well as by the frequency used. Heating time, power, power density (KW/in^2 of surface within the coil) and thermal conductivity of the part are thus important considerations. Further control of the heated depth can be obtained by scanning . . . i.e., moving the part through the coil progressively. (See pages 16 and 17)

APPLICATION OF PRINCIPLES

In Lepel high frequency induction heating equipment, 60 cycle current from the power line is passed through an electronic tube generator which changes it to high frequency current deemed most practical for industrial heating applications. A frequency of 450 kc/sec. is used for most metal treating applications, while megacycle frequencies of 2-40 megacycles/sec. are used for zone processing and crystal growing operations involving semiconductor and nonmetallic materials. The high frequency current thus generated is conducted through leads to induction coils in such a way as to minimize power losses.

BRAZING

In this process, the parts to be joined are assembled with the surfaces of the joints pre-fluxed and brazing alloy in the form of rings, strips, or irregular shapes preplaced at the joint area. Heat produced by induction coils rapidly heats the joint area to suitable temperature, melting the alloy and completing the joining operation. Heating is normally localized about the joint and requires but a few seconds, even on the most intricate assemblies. A wide range of alloys, on a silver-copper base, differing in color, melting temperature, fluidity and joint strength, meet the requirements of almost any assembly. Materials successfully joined by Lepel high frequency induction heating include standard carbon and alloy steels, brass, copper, stainless steel, cast iron, and many special ferrous, non-ferrous and refractory alloys. Frequently the joint is stronger than the material of the individual parts.

SOLDERING

Induction soldering has proved very effective for production applications where joint strength is not a primary consideration. The joining procedure is similar to that used in induction brazing, except that the use of lower joining temperatures increases production. Various solder alloys, primarily lead-tin base and a variety of fluxes including some which are non-corrosive, are available.

The design of both solder and braze joints is important for best results. Proper clearances, joint geometry, and provision for alloy preplacement influence joint strength and production handling. Best results are obtained when the parts to be joined are designed with induction heating in mind; however, good joints can often be achieved where induction heating was not contemplated in the original design.

SHRINK FITTING

Strong joints are effectively produced between parts of proper geometry by shrink fitting. In this process, one of the parts to be joined is generally expanded sufficiently by heating to slip into position over the other part and then allowed to cool, creating interference and thereby a shrink-fit joint. Such joints are produced with greatest simplicity at low enough temperatures to accommodate hardened parts and without the use of joining alloys. Moreover, this method is adaptable to parts made of dissimilar materials. Induction heating and proper fixturing provide shrink-fits at low production costs.

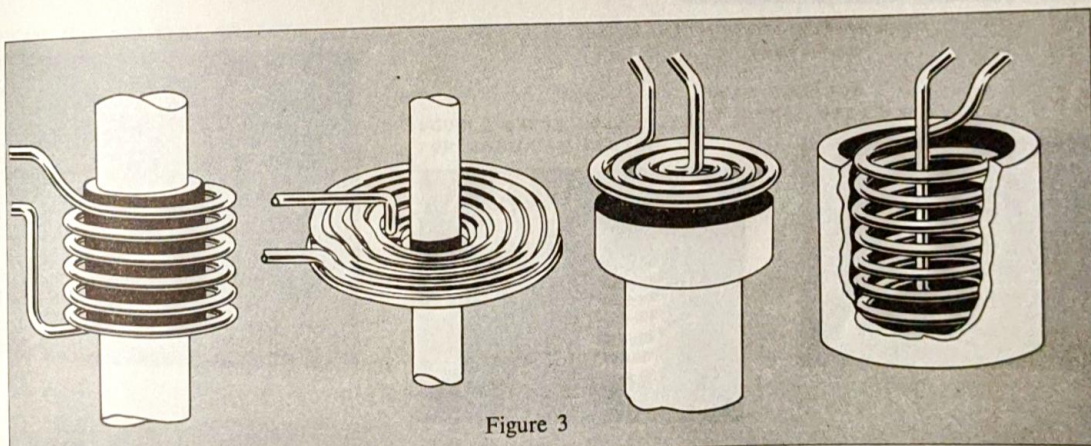
HEAT PATTERN

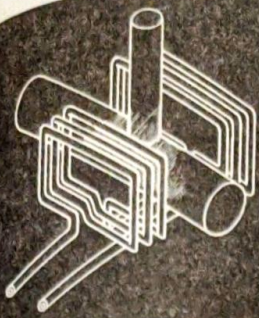
Ultimately, the success of a given induction heating application depends upon developing a suitable heat pattern in the part to be processed. This depends upon (1) the geometry of the part, (2) its electrical, magnetic and thermal properties, and (3) the coil design. Shown in Figure 3 are a variety of basic coil designs and the approximate heat patterns obtained. As indicated, heating may be highly localized, restricted to a given external or internal surface, on one end only, or in a desired geometric pattern. Since the induction coils for high frequency heating are made from copper tubing, they can be shaped to provide relatively complex heat patterns. The selective nature of induction heating is apparent from the illustrations.

INFLUENCE OF MATERIAL

Since any electrical conductor can be heated by induction, all ferrous and non-ferrous metals are effectively heated. However, electrical resistivity, magnetic permeability, specific heat and thermal conductivity variations may have a significant influence on heating rate and upon the heat pattern produced. Short heating cycles...i.e., 1 to 10 seconds...may require special consideration in processing operations which are temperature and time dependent; frequently higher temperatures compensate for the shorter times. Very pure semi-conductors and non-metallic materials of high electrical resistivity at room temperature often require use of megacycle frequencies and/or preheating before they can be heated directly by induction.

Any material can be indirectly heated using a conducting susceptor.





Load coil for soldering or brazing pipe joints.

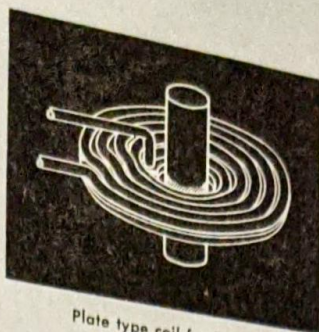
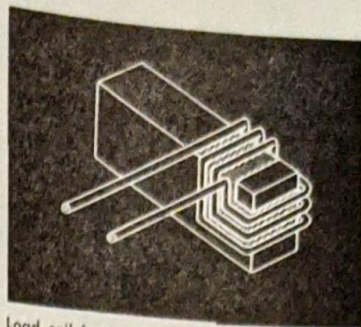


Plate type coil for heating a narrow band.



Load coil for brazing carbide tips to cutting tools.

LOAD COIL CONSTRUCTION

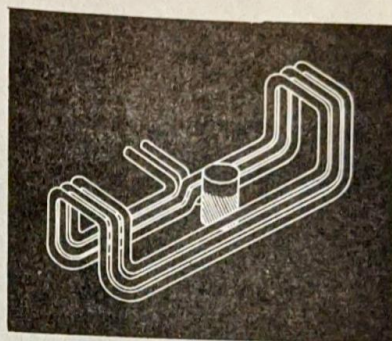
The illustrations on this page show a variety of basic work coil designs; from the simple, efficient helix type, to the more intricate coils. Coils of various shapes and sizes are embraced within each basic coil design.



Concentrator type load coil for extremely localized heating.

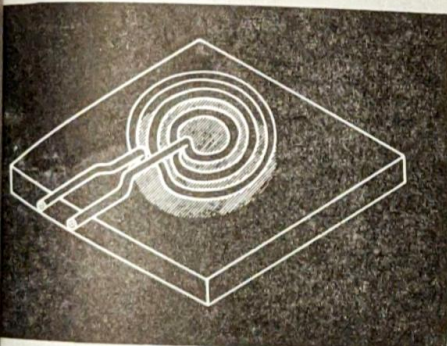
Coils are usually made from copper tubing, which can be readily shaped on hard wood forms with simple hand tools. Single loop cast or forged copper work coils are also used. Water, circulating through the coils, acts as a coolant when in operation.

The Lepel engineering laboratory is always available to assist you in selecting the most efficient load coil for your particular application.

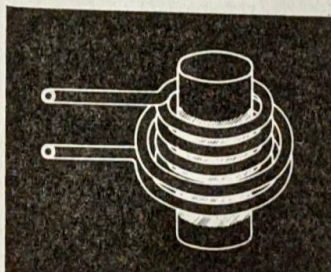


Conveyor type coil for continuous feed fixtures.

PANCAKE COIL FOR FLAT OBJECTS



DOUBLE LAYER LOAD COIL FOR GRADIENT HEATING



LOAD COIL FOR INTERNAL HEATING

